

## Technology Stack

|                      |                                        |
|----------------------|----------------------------------------|
| <b>Date</b>          | 8 July 2026                            |
| <b>Team ID</b>       |                                        |
| <b>Project Name</b>  | Credit Card Approval Prediction System |
| <b>Maximum Marks</b> | 2 Marks                                |

## Technology Stack Details

| <b>S.No</b> | <b>Architecture Component / Layer</b> | <b>Technology Chosen</b>                                                            | <b>Justification / Purpose</b>                                                                                                           |
|-------------|---------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | Frontend / Client-Side                | HTML5, CSS3, Bootstrap 5, JavaScript                                                | Used to build a responsive, user-friendly web interface for entering applicant details and displaying prediction results.                |
| 2           | Backend / Server-Side                 | Python 3, Flask                                                                     | Flask handles HTTP requests, processes user input, loads the trained ML model, and returns prediction results efficiently.               |
| 3           | Database / Data Storage               | CSV Dataset (application_record.csv & credit_record.csv), Joblib (.pkl model files) | Applicant data is stored in CSV files for training, while the trained machine learning model is saved as .pkl files for fast prediction. |
| 4           | Cloud / Hosting / Deployment          | IBM Watson Machine Learning, IBM Cloud, Render (or Local Flask Deployment)          | Enables cloud deployment, scalability, and real-time prediction through a web application.                                               |

| <b>S.No</b> | <b>Architecture Component / Layer</b> | <b>Technology Chosen</b>                                                 | <b>Justification / Purpose</b>                                                                                |
|-------------|---------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>5</b>    | <b>Version Control &amp; CI/CD</b>    | <b>Git, GitHub</b>                                                       | Used for source code management, version tracking, collaboration, and deployment integration.                 |
| <b>6</b>    | <b>Third-Party APIs / Other Tools</b> | <b>Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib</b> | Used for data preprocessing, visualization, model training, evaluation, and saving/loading the trained model. |